

Vortex-Induced Vibration Optimization of a Wide Streamline Box Girder by Wind Tunnel Test

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Abstract

Although the streamline box girder exhibits an excellent aerodynamic performance, they are often suffer severe Vortex-Induced Vibrations (VIVs) due to complicated separation and reattachment of air flows. It is necessary to apply control measures on the streamline box girder to suppress VIV. In this study, the VIV optimizations of a super wide streamline box girder were conducted via a series of section model wind tunnel tests. The effects of wind fairings, inspection vehicle rails, guide vanes, traffic barriers and handrails on the heaving VIV performance of the wide streamline box girder were analyzed. The test results show that the inspection vehicle rail and handrail are the main members to induce the heaving VIV of the box girder. A sharper wind fairing without changing the distance of railings is favorable to the heaving VIV performance. However, the countermeasures of changing the position of inspection vehicle rail and installing guide vanes under the girder have slight influences on reducing heaving VIV responses. Inspired by the spanwise sinusoidal perturbation method, the traffic barriers sealed with plates by regular intervals along the bridge deck can suppress the heaving VIV of the box girder successfully. The porosity of handrails has a significant effect on the heaving VIV. By using a sharp wind fairing and circular simplified handrails, the VIV performance of the wide streamline box girder was greatly improved. Their mitigation abilities were verified by a large-scale section model wind tunnel test.

Keywords: wide streamline box girder; vortex-induced vibration; aerodynamic countermeasure; wind tunnel test

1. Introduction

Owing to excellent mechanical properties and aerodynamic characteristics, the streamline box girders have been extensively used in modern long-span bridges. These bridges include Xihoumen Suspension Bridge (main span: 1650 m), Great Belt East Suspension Bridge (main span: 1624 m), Yi Sun-sin Suspension Bridge (main span: 1545 m), Stonecutters Cable-Stayed Bridge (main span: 1018) and Russky Island Cable-Stayed Bridge (main span: 1104). The bridges become more flexible, lower in inherent damping and light in weight with the increase in span-length, which may cause a susceptibility to the wind excitations.

As a common aeroelastic phenomenon, vortex-induced vibration (VIV) may occur in flexible structures when subjected to wind actions and it becomes a major challenge in the design of long-span bridges. VIV have been observed in many real bridges at low wind velocities, such as Great Belt East Bridge in Denmark (Larsen *et al.*, 2000; Frandsen, 2001), Xihoumen Bridge in China (Li *et al.*, 2014; Yang *et al.*, 2015), Rio-Niterói Bridge in Brazil (Battista and Pfeil, 2000) and Trans-Tokyo Bay Crossing

Bridge in Japan (Fujino and Yoshida, 2002). Large VIV oscillations will not only lead to discomfort of drivers but also long-term fatigue damages in structures. It is therefore necessary to use effective countermeasures to mitigate VIV and attenuate VIV responses under the allowable values. Larsen *et al.* (2000) experimentally studied the VIV of Great Belt East Bridge and using the guide vanes that installed at the joints of bottom and side panel to suppress the vortex shedding responses. Ge *et al.* (2011) found that a good VIV control effect of Xihoumen Suspension Bridge can be achieved by installing adjustable wind barriers. Battista and Pfeil (2000) investigated the effectiveness of Tuned-Mass Dampers (TMDs) on attenuating the vortex-induced oscillations of Rio-Niterói Bridge. The performances of TMDs were analyzed through the comparison of numerical results obtained for time responses of the original and the controlled bridge. Fujino and Yoshida (2002) developed the TMDs specifically to control the first and the second vertical flexural modes of Trans-Tokyo Bay Crossing Bridge and found that TMDs exhibited sufficient vibration control.

Generally, the most commonly used control measures for the

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suppression of VIV of long-span bridges are aerodynamic countermeasures. The VIV aerodynamic countermeasures of streamline box girders have been extensively studied and applied to many long-span bridges. Larsen *et al.* (2008) studied the effects of guide vanes on the vortex shedding responses of a twin box girder at three different Reynolds numbers. The test results show that it becomes important to ensure sufficiently high Reynolds numbers to avoid thick boundary layers growing on the model and thus prevent high flow rates into the guide vane. Sun *et al.* (2012) studied the effects of handrail, inspection vehicle rail and guide vane on the VIV of a closed box girder by large-scale section model wind tunnel tests. The test results indicate that the VIV could be mitigated by setting guide vane inside the inspection vehicle rail. By section model and full bridge aeroelastic model wind tunnel tests, Sun *et al.* (2013) investigated the VIV performance of a box girder and discussed the differences of the results between section and full aeroelastic model testing results. The studies demonstrate that a good consistency between section model and full bridge model can be achieved by considering the spanwise correlations of vortex-induced forces. Xu *et al.* (2016) explored the torsional VIVs mechanism of box girder deck by performing simultaneous pressure measurements. The results indicate that the angle of incidence, flow conditions and installation of a spoiler exert significant effects on the surface pressure distributions. The potential torsional VIVs could be suppressed by the interaction of vortices generated by the spoiler with an appropriate configuration and location. El-Gammal *et al.* (2007) conducted an experimental study to determine the effectiveness of the Spanwise Sinusoidal Perturbation Method (SPPM) in controlling VIV in a plate girder bridge section model. The results show that the VIV could be successfully reduced by the SPPM method. Yang *et al.* (2014) experimentally studied the effectiveness of guide vane, grid plate and wind barrier on

mitigating VIV of three different types of twin-box girders and found that the wind barrier is the most effective countermeasure. Laima *et al.* (2018) carried out an experimental investigation on the effects of attachments on the aerodynamic characteristics and VIV of a twin-box girder. The flow pattern, pressure distribution, vortex shedding frequency and VIV of five models with different attachments were compared with a bare deck. The results demonstrate that the flow characteristics and VIVs of the twin-box girder with attachments differ considerably from the bare twin-box girder owing to the large leading flow separation induced by the attachments.

The main purpose of the present study is to investigate the VIV performance of a super wide streamline box girder and explore effective VIV aerodynamic countermeasures. Optimization wind tunnel tests were performed to study the mitigation effects of wind fairings, inspection vehicle rails, guide vanes, traffic barriers and handrails on the VIV performance of the wide streamline box girder. A large-scale section model wind tunnel test was conducted to further verify the VIV performance and the mitigation effects of the proposed optimal countermeasure.

2. Engineering Background

The Qingshan Yangtze River Bridge, located in Wuhan City of China, is a long-span cable-stayed bridge crossing the Yangtze River. The bridge has seven spans of 100 + 102 + 148 + 938 + 148 + 102 + 100 m. The double fan-shaped cable planes are adopted in the bridge. The two towers are reinforced concrete structures with an overall height of 271.5 and 279.5 m, respectively. The typical streamline box girder is shown in Fig. 1(b). The bridge has the widest single box girder in the world, the width of the bridge deck is 47.0 m, and the height is 4.5 m. Owing to such a large width of the streamline box

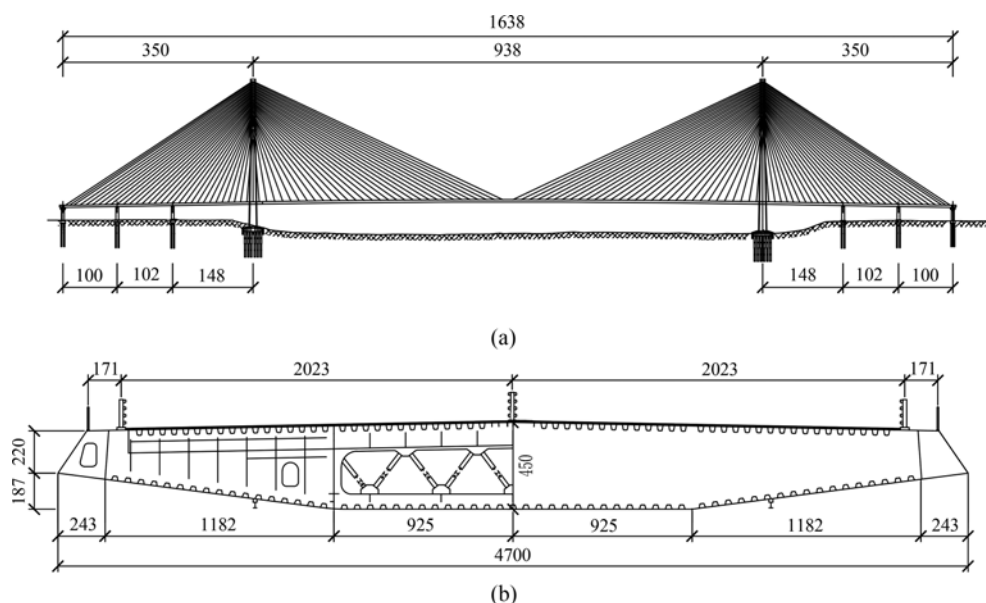


Fig. 1. Schematic Description of Qingshan Yangtze River Bridge: (a) Elevation (Unit: m), (b) Box Girder Section (Unit: cm)

girder, the VIV may occur in a low wind velocity for the flexible long-span bridge. Therefore, it is essential to study the VIV performance of the wide streamline box girder and propose effective VIV countermeasures.

3. Wind Tunnel Test

3.1 Experimental Set-up

A series of spring-suspended section model tests were conducted to investigate the VIV performance and find effective VIV aerodynamic countermeasures of the wide streamline box girder. The section models involving 1:50 and 1:27 scale ratios, as shown in Fig. 2. To investigate the VIV performance and the mitigation effects of countermeasures of the box girder, 1:50 scale ratio section model wind tunnel tests were carried out in the XNJD-1 Wind Tunnel, a closed-circuit-type wind tunnel with a work section 2.4 m (width) $\times$ 2 m (height) in size. The wind speed is adjustable continuously from 0.5 to 45.0 m/s and the turbulent intensity is less than 0.5% at the concerned wind speeds. To verify the VIV performance and the mitigation effects of the recommended countermeasures, 1:27 scale ratio section model wind tunnel tests were conducted in the XNJD-3 Boundary Layer Wind Tunnel, which is a closed circuit type with a height, width and length of 4.5, 22.5 m and 36 m, respectively.

The main body of the section models were made of wooden materials. The railings and other subsidiary facilities on decks

were carved by Acrylonitrile Butadiene Styrene (ABS) material precisely to ensure excellent simulations of aerodynamic configurations. According to the *Wind-resistant design specification for highway bridges* (CMT, 2004), the aspect ratio (the ratio of length to width) of section model should be larger than 2.0 if the test is conducted in a closed type wind tunnel. In the present study, to have precise simulations of ancillary facilities of bridge deck, the aspect ratios of 1:50 and 1:27 section models are respectively chosen as 2.2 and 2.0 in the premise of satisfying the wind resistance specification requirements and the wind tunnel dimensions. In addition, to ensure a two-dimensional (2D) flow and reduce the effects of the bar of support system and the end effects of section models, two wooden end plates and two rigid streamlined temporary end walls were used in the conventional and the large-scale section model tests, respectively.

In the tests, the section models were elastically suspended by four pairs of pre-tensioned linear springs, which can simulate the bending and torsional modes in two degrees of freedom. Three displacement laser sensors were used to monitor the vibrations of models. The sampling frequency was set at 256 Hz for a sampling rate of 60 seconds. Table 1 lists the structural parameters including the geometric dimensions, the mass characteristics, the fundamental frequencies and the structural damping of the 1:50 and 1:27 scale models.

Although the damping ratio has a dramatic effect on VIV, there has not been a widely accepted method to estimate the damping



Fig. 2. Section Models in Wind Tunnels: (a) 1:50 Scale Section Model in XNJD-1 Wind Tunnel, (b) 1:27 Scale Section Model in XNJD-3 Boundary Layer Wind Tunnel

Table 1. The Parameters of Section Model Wind Tunnel Tests

Properties		Prototype	Section models (scale ratio)	
			1:50	1:27
Geometric scale	Length L (m)	–	2.095	3.460
	Width B (m)	47.0	0.940	1.741
	Height H (m)	4.5	0.090	0.167
Equivalent mass	Mass per unit length m (kg·m ⁻¹)	47 158	18.863	64.688
	Mass moment of inertia per unit length I_m (kg·m)	7 777 960	1.244	14.639
Frequency	Vertical f_h (Hz)	0.213	2.873	3.353
	Torsional f_t (Hz)	0.559	7.342	6.754
Damping ratio	Vertical ζ_h (%)	–	0.362	0.373
	Torsional ζ_t (%)	–	0.351	0.306

ratios of bridge structures at present. It is recommended to be taken as 0.5% based on CMT (2004). While according to *the Design manual for roads and bridges* (HAE, 2001), the damping ratio should be taken as 0.36% for steel bridges. The previous field measurement results show that VIVs may occur in a lower damping ratio condition. In addition, to ensure the safety of the bridge and make the VIV responses easily observed, the damping ratio of the section model was selected as 0.362% in the optimization tests, which is very close to the suggestive value of the British specification (HAE, 2001). Meanwhile, the effects of damping ratio on the VIV responses of original deck and the optimal VIV countermeasure were studied in three different damping ratio levels.

3.2 VIV Performance of Original Deck

For the wide streamline box girder with the original section, Figs. 3-4 respectively show the VIV responses of heaving and torsional standard deviations at different attack angles in smooth flow, in which the VIV responses and the wind velocities have been transferred to the prototype values. The heaving VIVs were observed and there are two lock-in regions under $+3^\circ$ and $+5^\circ$ attack angles. The maximum amplitudes of the first and the second lock-in regions are 0.107 and 0.135 m under $+3^\circ$ attack

angle, 0.074 and 0.296 m under $+5^\circ$ attack angle, respectively. The torsional VIVs were observed and the maximum amplitudes are 0.042° , 0.061° , 0.187° and 0.259° under -5° , 0° , $+3^\circ$ and $+5^\circ$ attack angles, respectively.

According to HAE (2001), dynamic sensitivity parameters K_D for the bending and torsional effects are respectively defined as follows:

$$K_D = y_{\max} f_h^2 \quad (1)$$

$$K_D = y_{\max} f_t^2 \quad (2)$$

where $y_{\max}$ is the predicted bending or torsional amplitude. If the critical wind speed for the excitation of the relevant mode is less than 20 m/s and K_D is greater than 30 mm/s^2 , pedestrian discomfort may be experienced and the VIV mitigation countermeasures should be proposed. Based on Eqs. (1)-(2), the maximum standard deviations calculated from the allowable acceleration ($K_D = 5 \text{ mm/s}^2$) are 0.661 m and 0.234° for the fundamental modes of heaving and torsional VIVs, respectively.

Based on the Chinese wind resistant specification (CMT, 2004), the amplitude limitations of the fundamental modes of heaving and torsional VIVs ($[h_a]$ and $[\theta_a]$) are specified directly, they are respectively defined as follows:

$$[h_a] = 0.04/f_h = 0.188 \text{ m} \quad (3)$$

$$[\theta_a] = 4.56/(Bf_t) = 0.174^\circ \quad (4)$$

Compared with the above VIV amplitude limitations, the maximum amplitudes of the heaving VIV under $+5^\circ$ attack angle and the torsional VIV under $+3^\circ$ and $+5^\circ$ attack angles cannot satisfy the design requirements. With the consideration of the velocities of the lock-in regions of the torsional VIVs are nearly 30 m/s, the pedestrian discomfort problems mainly induced by buffeting responses, the torsional VIVs can thus be neglected due to such high wind velocities. However, the wind velocity corresponding to the maximum heaving VIV amplitude under $+5^\circ$ attack angle is approximately 19.3 m/s. In addition, the meteorological parameters near the bridge site show that large attack angles exist occasionally. The flow conditions of the bridge site are somewhat complicated and uncertain. With considerations to such low wind speed and the complicated surround terrains at bridge site, the heaving VIV of the wide streamline box girder under $+5^\circ$ attack angle should therefore be optimized.

As mentioned above, the damping ratio might have dramatic effects on VIV responses. To study the effect of damping ratio experimentally, wind tunnel tests were performed on the original deck at $+5^\circ$ attack angle in three different damping ratio levels (for heaving VIV) including, 0.272%, 0.362% and 0.508%. The peak values of RMS of heaving VIV responses in the first and the second lock-in regions under above three damping levels are shown in Fig. 6. The results show that the RMS values in the first and the second lock-in regions decrease inversely with the increase of damping ratio, which agrees with the conclusion proposed by Zhu (2005). It should be noted that

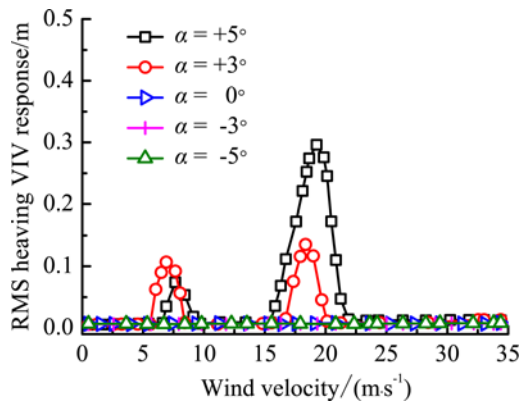


Fig. 3. Heaving VIV Responses of the Girder with the Original Section

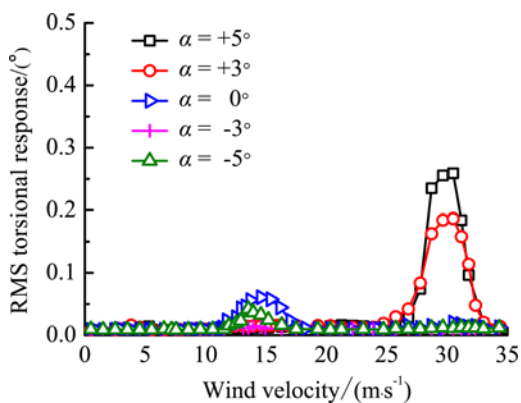


Fig. 4. Torsional VIV Responses of the Girder with the Original Section

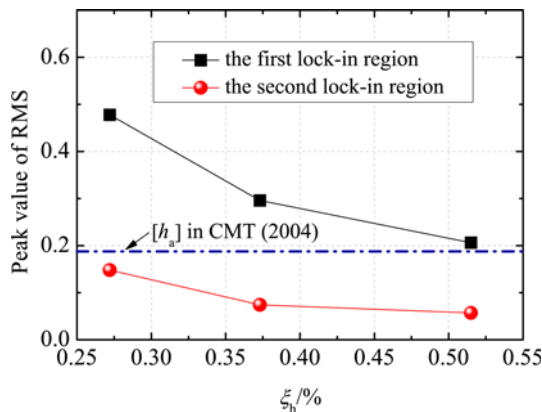


Fig. 5. The Peak Values of RMS of Heaving VIVs Vary with the Damping Ratios at +5° Attack Angle

the peak value of RMS in the second lock-in region is still larger than the amplitude limitation when the damping ratio is 0.508%. Effective aerodynamic countermeasures should be proposed to improve the heaving VIV performance of the wide streamline box girder.

4. Aerodynamic Countermeasures

To suppress the heaving VIV of the wide streamline box girder, various aerodynamic countermeasures were studied and their effects on the suppression of VIV were analyzed via 1:50 section model wind tunnel tests. The most unfavorable wind attack angle +5° was chosen to perform the optimizing tests and the test parameters were identical to the parameters listed in Table 1. The aerodynamic countermeasures include the modification of aerodynamic configurations of the deck and using accessional mitigation measures. The effects of wind faring, inspection vehicle rail, guide vane, traffic barrier and handrail on the heaving VIV displacement were analyzed and the effective optimization schemes were found in the tests.

4.1 The Effect of Wind Faring

The aerodynamic configurations of bridge deck have significant effects on the VIV performance and it is possible to shape bridge deck sections that vortex shedding excitations can be suppressed for practical levels of structural damping (Larsen and Wall, 2012). As an important part of aerodynamic configuration of box girder, the wind faring has a dramatic effect on the aerodynamic characteristics of main girder. Meng *et al.* (2011) investigated the influence of wind faring angles on the VIV and flutter performances of box girders via wind tunnel tests and Computational Fluid Dynamics (CFD). The results indicate that the VIV performance could be improved via shaping wind fairings without lowering the flutter stability. Qian *et al.* (2015) studied the VIV and control measures for the Π shaped bridge deck and found that the VIV can be mitigated by sharpening the wind fairings.

As shown in Fig. 6, the wind fairings in different shapes were investigated based on the previous studies. Case 0 shows the

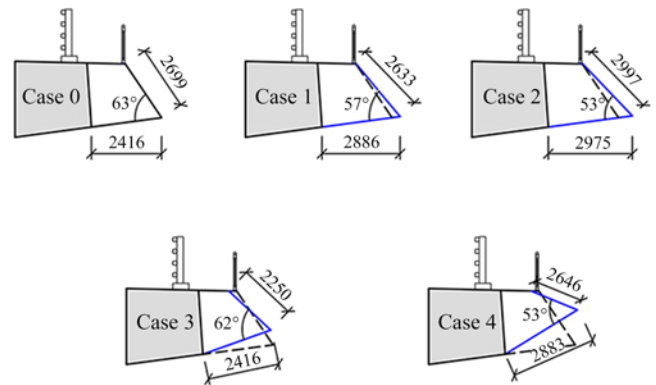


Fig. 6. Wind Fairings in Cases 0-4 (unit: cm)

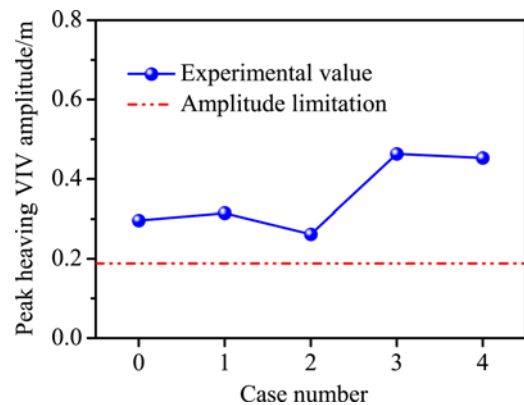


Fig. 7. The Effect of Wind Faring on the Peak Amplitudes of Heaving VIV Displacements

wind faring of original deck. In Cases 1-2, the angle of wind faring is changed. In Cases 3-4, both the angle of wind fairings and the distance between the railings are changed. The peak amplitudes of heaving VIV displacements in Cases 0-4 are shown in Fig. 7.

It can be seen that the sharp wind faring in Case 2 has a better mitigation effect than other cases, which can reduce the heaving VIV amplitude by 11.5% compared with the result of original deck. However it cannot satisfy the design requirement. Compared with the bluff wind fairings, a sharp wind faring can reduce the flow separation at the leading edge of box girder. The vortex shedding thus weakens and the VIV performance is improved. However, the peak heaving VIV displacement is still higher than the allowable amplitude. Compared with the test results of Case 2 and Case 4, it can be found that a narrow distance between the railings on the deck has an adverse effect on the VIV performance. This indicates that a close distance of railings exacerbates the flow separation and it is detrimental to the heaving VIV of wide streamline box girder. Owing to a better aerodynamic performance, the modified section model with the wind faring used in Case 2 was adopted in the following optimizing tests.

4.2 The Effect of Inspection Vehicle Rail

Apart from the wind fairing, the inspection vehicle rails are also important structural components to the aerodynamic configurations of bridge deck, which can induce large vortex shedding and have a considerable influence on the VIV performance of box girder (Sun *et al.*, 2012).

Table 2 lists the countermeasures of the inspection vehicle rails. In Case 5, the inspection vehicle rails are abandoned to investigate their effects on the heaving VIV. In Cases 6-9, the inspection vehicle rails are installed on the inclined side panels and the installing positions are changed. In Cases 10-13, the inspection vehicle rails are installed on the box bottom plates and both the installing positions and the height of supports are changed. The peak amplitudes of heaving VIV displacements in Cases 5-13 are shown in Fig. 8.

The results show that the heaving VIV is significantly suppressed when the inspection vehicle rails are abandoned. The heaving VIV amplitude is reduced by 64.2% compared with the result of original deck. This indicates that the inspection vehicle rails are the main structural members to cause the heaving VIV. The peak amplitudes of VIV decrease gradually as the inspection vehicle rails installed on the inclined side panel moving outwards of box girder. A better mitigation effect can be achieved when the inspection vehicle rails have a lower height and moving inside of the bottom plate of box girder. However, there are slight effects on the improvement of heaving VIV performance by

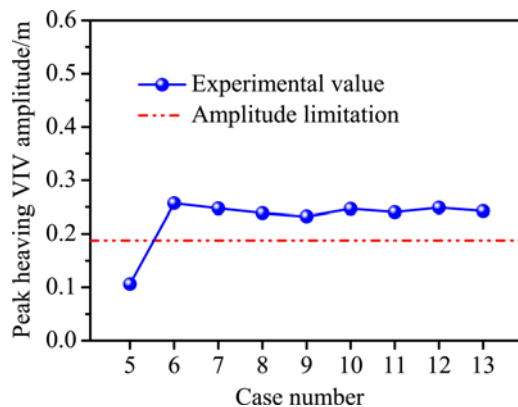


Fig. 8. The Effect of Inspection Vehicle Rail on the Peak Amplitudes of Heaving VIV Displacements

changing the installation positions and the height of supports of the inspection vehicle rails.

4.3 The Effect of Guide Vane

Guide vane can smooth the air flow around the corner of bridge deck and the effectiveness of guide vane is closely related to the installation positions and dimensions. (Larsen *et al.*, 2008). The VIVs of the Great Belt East Bridge were mitigated efficiently by fitting guide vanes at the joints of the horizontal bottom plate and the lower side panels of box girders. The guide vanes can introduce high-speed flows in the wake region of the upwind box and thus preventing the vortex formation (Larsen *et al.*, 2000).

The effects of the installation position, the inclined angle and the length of guide vanes on heaving VIVs were investigated in the tests. Table 3 lists the countermeasures of the guide vanes. In Cases 14-17, the guide vanes are installed beside the inspection vehicle rails on the inclined side panels. In Cases 18-19, the guide vanes are installed beside the inspection vehicle rails on the bottom plate. In Cases 20-22, the guide vanes are installed in the corner of box girder, and the inspection vehicle rails are installed on the inclined side panels. In Cases 23-24, the guide vanes are installed in the corner of box girder, and the inspection vehicle rails are installed the bottom plate. The support height of the inspection vehicle rail is 0.48 m. The results of the peak amplitudes of heaving VIV displacements are shown in Fig. 9.

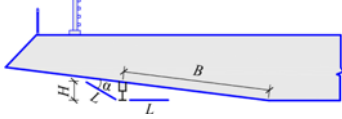
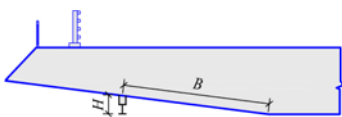
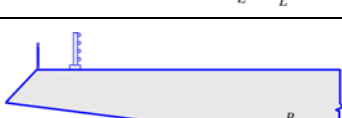
It can be seen that a long guide vane has a better mitigation effects than the short ones. For the guide vanes installed beside the inspection vehicle rails, a smaller slope between the guide vane and box girder is benefited to the heaving VIV to some extent. For the guide vanes installed in the corner of box girder, the peak amplitudes of VIV decrease when the inspection vehicle rails installed on the inclined side panel moving outwards or the inspection vehicle rails installed on the bottom plate moving inwards of the box girder. Although by installing guide vanes in different positions have positive effects on reducing heaving VIV responses in a varying degree, the peak heaving VIV amplitudes are reduced by no more than 30.1% and they are still higher than the amplitude limitation. Therefore, the installation of guide vanes cannot effectively suppress the heaving VIV of

Table 2. Countermeasures of the Inspection Vehicle Rail

Countermeasures	Cases and description
No inspection vehicle rail*	Case 5
Inspection vehicle rails are installed on the inclined side panel of box girder	Case 6: $B = 4.6$ m, $H = 0.48$ m Case 7: $B = 5.2$ m, $H = 0.48$ m Case 8: $B = 6.0$ m, $H = 0.48$ m Case 9: $B = 7.2$ m, $H = 0.48$ m
Inspection vehicle rails are installed on the bottom plate of box girder	Case 10: $B = 2.4$ m, $H = 0.48$ m Case 11: $B = 3.0$ m, $H = 0.48$ m Case 12: $B = 2.4$ m, $H = 0.70$ m Case 13: $B = 3.0$ m, $H = 0.70$ m

*the original inspection vehicle rails are installed on the inclined side panel, $B = 4.1$ m, $H = 0.48$ m.

Table 3. Countermeasures of the Guide Vanes

Countermeasures		Cases and description
Guide vanes are installed beside the inspection vehicle rails on the inclined side panel of box girder		Case 14: $B = 4.1$ m, $L = 2.2$ m, $\alpha = 18^\circ$ Case 15: $B = 4.1$ m, $L = 3.0$ m, $\alpha = 15^\circ$ Case 16: $B = 7.2$ m, $L = 2.2$ m, $\alpha = 18^\circ$ Case 17: $B = 7.2$ m, $L = 3.0$ m, $\alpha = 15^\circ$
Guide vanes are installed beside the inspection vehicle rails on the bottom plate of box girder		Case 18: $B = 2.2$ m, $L = 2.2$ m, $\alpha = 13^\circ$ Case 19: $B = 3.1$ m, $L = 3.0$ m, $\alpha = 9^\circ$
Guide vanes are installed in the corner of box girder, the inspection vehicle rails installed on the inclined side panel of box girder		Case 20: $B = 4.1$ m, $L = 3.0$ m, $\alpha = 173^\circ$ Case 21: $B = 7.2$ m, $L = 3.0$ m, $\alpha = 173^\circ$ Case 22: $B = 7.2$ m, $L = 2.2$ m, $\alpha = 173^\circ$
Guide vanes are installed in the corner of box girder, the inspection vehicle rails installed on the bottom plate of box girder		Case 23: $B = 2.25$ m, $L = 2.2$ m, $\alpha = 173^\circ$ Case 24: $B = 3.05$ m, $L = 3.0$ m, $\alpha = 173^\circ$

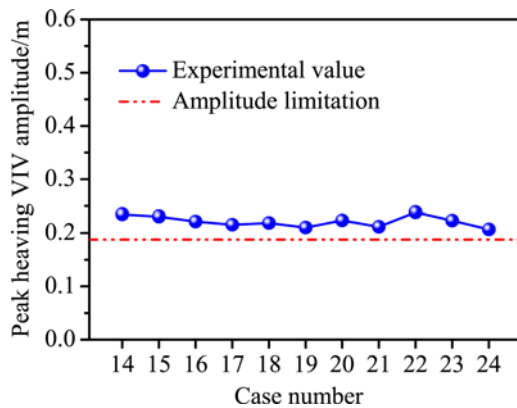


Fig. 9. The Effect of Guide Vane on the Peak Amplitudes of Heaving VIV Displacements

the wide streamline box girder. Other effective aerodynamic countermeasures are still needed to explore to improve the VIV performance.

4.4 The Effect of Traffic Barrier

The Spanwise Sinusoidal Perturbation Method (SPPM) that introducing three-dimensional periodic geometric disturbances on wake generators is an efficient method in controlling VIV by reducing the spanwise correlations of VIV forces (Owen *et al.*, 2001; Bearman and Branković, 2004, El-Gammal *et al.*, 2007). Inspired by the SPPM, the traffic barriers were sealed with plates by regular intervals along the bridge deck, as shown in Fig. 10. Five different intervals of plates ($n = 1, 2, 3, 5$ and 7) were used to seal the traffic barriers and the corresponding porosities (the air permeability per unit area) of the traffic barrier are 24.6%, 32.8%, 36.9%, 41.0% and 43.1% in Cases 25-29, respectively.

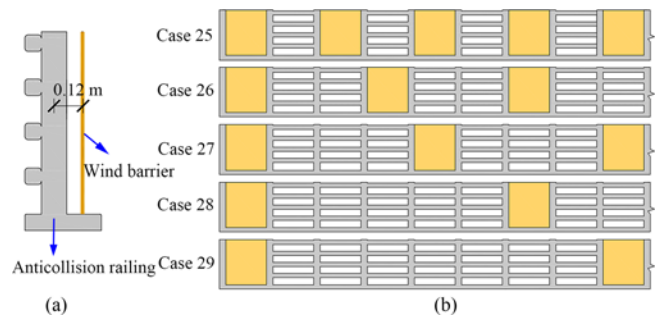


Fig. 10. Schematic Diagram of Traffic Barrier and Arrangements in Cases 25-29

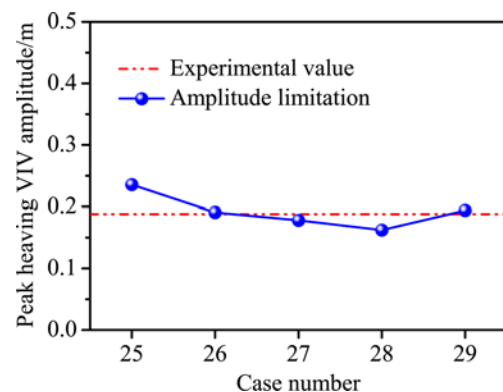


Fig. 11. The Effect of Traffic Barrier on the Peak Amplitude of Heaving VIV Displacements

The results of the peak amplitude of heaving VIV displacements are shown in Fig. 11.

The test results show that the traffic barrier in 41.0% porosity (interval of plates, $n = 5$) can achieve a good mitigation effect.

Compared with the results of original deck, the heaving VIV amplitude is reduced by 45.3%, which can satisfy the design requirements. The test results demonstrate that the traffic barriers sealed with plates by regular intervals along the bridge deck could affect the flow characteristic and vortices formation. The spanwise correlations of heaving VIV forces are weakened. The heaving VIV is thus greatly improved by sealing the traffic barriers at intervals along the bridge deck.

It is noteworthy to say that, although sealing the traffic barrier at intervals is an effective countermeasure to suppress the heaving VIV, it would increase the drag force of wind loading of the streamline box girder. Furthermore, the plate installed behind the traffic barrier will not only block the field of driving vision but also affect the appearance of the urban bridge negatively. Therefore, sealing the traffic barrier is not the optimal aerodynamic countermeasure and it is essential to find other effective VIV countermeasures that have slight influences on the other properties of long-span bridges.

4.5 The Effect of Handrail

Because handrails are installed on the leading edge of bridge decks, the boundary layer is thus severely distributed and there are significant effects on the VIV performances of bridge decks. Matsumoto *et al.* (1993) experimentally studied the effects of the porosity of handrail on the vortex-induced oscillations of a hexagonal box girder in both smooth flow and turbulent flow. The test results indicate that the VIV amplitude shows a maximum when the porosity of handrail is 30%. Nagao *et al.* (1997) investigated the effects of a simplified handrail on the wind-induced oscillations of a box girder by wind tunnel tests and found that both the height and the shape of handrails have effects on the VIV amplitudes.

The effects of the porosity and the shape of handrails on the heaving VIV performance of wide streamline box girder were analyzed in the tests. To study the influence of handrail, the rectangular handrails are abandoned in Case 30. In Case 31, the porosity of handrails increases to 66% (the porosity of the handrail of the original deck is 56%) without changing the railing shape. In Case 32, the circular simplified handrails are used and the corresponding porosity further increases to 73%. The handrails

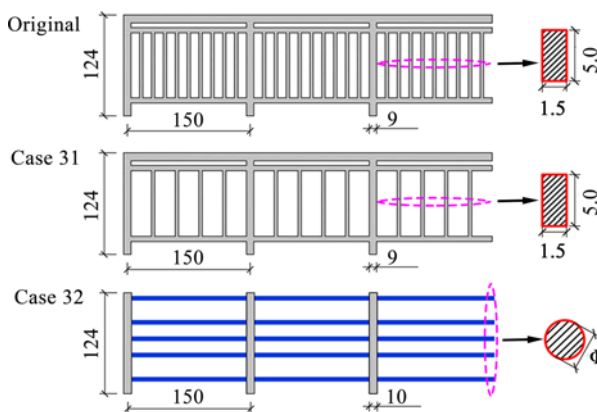


Fig. 12. Schematic Diagrams of Handrails (Unit: cm)

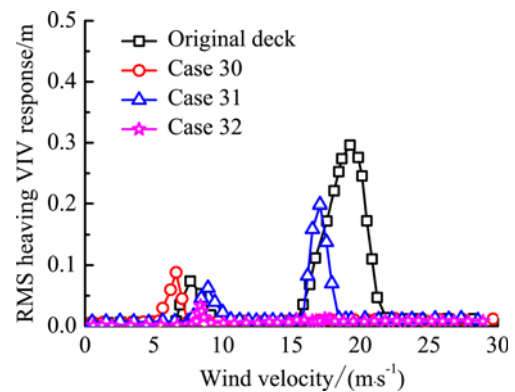


Fig. 13. The Effects of Handrails on the RMS Heaving VIV Responses

of the original deck and the handrails used in Cases 31-32 are shown in Fig. 12.

Figure 13 shows the RMS of the heaving VIV responses of the box girder with different handrails under $+5^\circ$ attack angle. Compared with the results of original deck, the second lock-in region of the heaving VIV disappears without the handrail. Furthermore, the peak amplitude of the first lock-in region decreases dramatically, the heaving VIV amplitude is reduced by 70.3% compared with the original deck. The results indicate that, besides the inspection vehicle rail, the handrail is another key factor to induce the heaving VIV. In Case 31, there are still two lock-in regions of heaving VIV and the peak amplitude reduced with the increase of the porosity of handrail. In Case 32, the second lock-in region of heaving VIV disappears and the peak VIV amplitude decreases to 0.034 m.

Compared with the test results of original deck, the countermeasure of the simplified handrail with circular section fundamentally change the VIV properties. Thus, the VIV responses of the deck with the circular simplified handrails under different attack angles are essential to be examined. The heaving VIV displacements of main girder with circular simplified handrails under different attack angles are shown in Fig. 14. The test results show that the heaving VIV amplitude under $+3^\circ$ attack angle is 0.102 m, which is larger than that of $+5^\circ$ attack angle

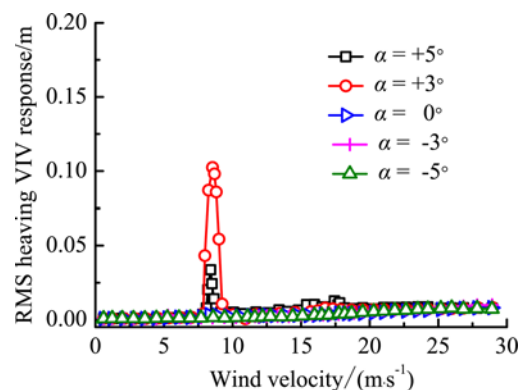


Fig. 14. Heaving VIV Response of the Girder with the Circular Simplified Handrails

with the value of 0.034 m. Correspondingly, the heaving VIV amplitude is reduced by 65.5% compared with the results of original deck. The VIV performances of the wide streamline box girder under different attack angles can satisfy the design requirements. The circular simplified handrail in a high porosity can permit the higher wind velocity passing through the deck surface and reduce the separations and reattachments of airflows, which can thus improve the heaving VIV performance significantly. Therefore, the circular simplified handrail used in Case 32 is regarded as the optimal aerodynamic countermeasure.

To ensure the safety of the long-span bridge, the deck with the simplified handrail was examined in the low damping ratio level. In addition, the VIV performance of the deck with the optimal countermeasure was also investigated in the high damping level for comparison. The peak values of RMS of heaving VIV are 0.149 m and 0.074 m in cases of 0.272% and 0.508% damping ratio, respectively. The wide streamline bridge deck with the simplified handrail therefore exhibits a good VIV performance even in a low damping ratio condition.

5. Verification of VIV Performance and Mitigation Effects

According to the results of previous studies (Larsen *et al.*, 2008; Yang *et al.*, 2016), a conventional model scale from 1:50 to 1:80 may not be able to simulate the aerodynamic characteristics of bridge deck accurately. Compared with the conventional scale section model tests, a large-scale section model test can probably provide a reasonable and effective prediction on VIV responses (Yang *et al.*, 2015). A 1:27 large-scale section model test was therefore conducted to verify the VIV performance and mitigation effects of the optimal countermeasure.

Firstly, the VIV characteristics of original deck were investigated under different wind attack angles. The test results of the heaving VIV responses are shown in Fig. 15. To compare the results of section model in two different scales, the heaving VIV responses under $+3^\circ$ and $+5^\circ$ attack angles are also included in Fig. 14. It can be seen that the heaving VIV occurred under $+3^\circ$ and $+5^\circ$ attack angles and there are two lock-in regions. The maximum amplitudes of the first and second lock-in regions are 0.082 and 0.126 m under $+3^\circ$ attack angle, 0.053 and 0.260 m under $+5^\circ$ attack angle, respectively. Comparing with the results of the 1:50 scale ratio section model test, the heaving VIV displacements of the 1:27 scale section model under $+3^\circ$ and $+5^\circ$ attack angles are slight smaller than those of the 1:50 scale section model. The same comparison results regarding the VIV responses of different scale ratios can be found in the previous studies (Zhang *et al.*, 2008; Yang *et al.*, 2016). This phenomenon may be attributed to the effects of Reynolds numbers. Nevertheless, the testing results are generally in a good agreement and the VIV performance of the original deck is verified.

From the results of the 1:50 scale section model test, the circular simplified handrail in high porosity is the most effective control countermeasure. More importantly, unlike the countermeasure

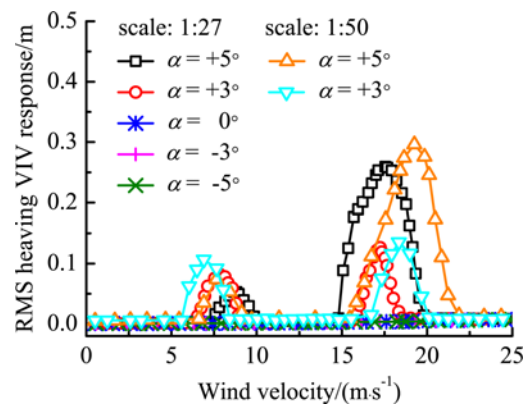


Fig. 15. Heaving VIV Responses of Original Deck by 1:27 Scale Section Model Test

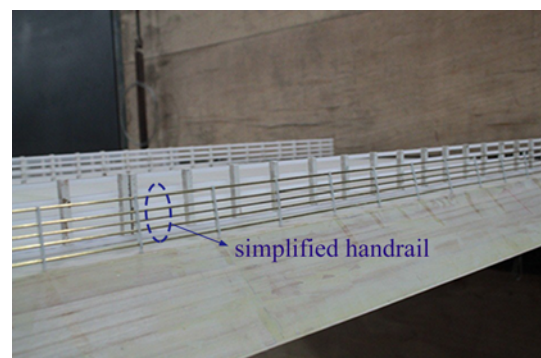


Fig. 16. Simplified Handrail on the 1:27 Scale Section Model

of the traffic barrier used in Cases 25-29, the simplified handrails have slight negative influences on the aerostatic wind loads, driving comfort and bridge aesthetics. Therefore, the mitigation effects of the simplified handrails on the VIV characteristics of the wide streamline box girder were investigated by a large-scale section model test. The simplified handrail with circular section on the 1:27 scale section model is shown in Fig. 16.

The test results of heaving and torsional VIV responses are shown in Figs. 17-18. It can be seen that the heaving VIV occurs under $+3^\circ$ and $+5^\circ$ attack angles, and the maximum amplitudes

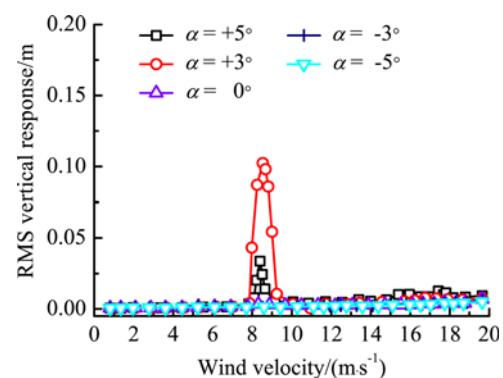


Fig. 17. Heaving VIV Response of the Box Girder with the Circular Simplified Handrail

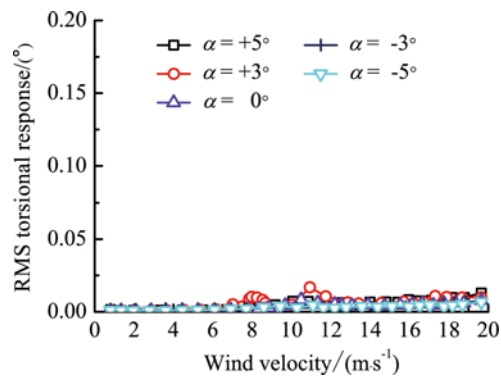


Fig. 18. Torsional VIV Response of the Box Girder with the Circular Simplified Handrail

of heaving VIV are 0.102 m and 0.034 m under $+3^\circ$ and $+5^\circ$ attack angles, respectively, which are smaller than the permissible value. Furthermore, no large torsional VIV responses occur under different attack angles in the tests. Therefore, the box girder exhibits a good VIV performance and can satisfy the wind-resistant requirements using the circular simplified handrail.

6. Conclusions

In this study, the VIV performance and various countermeasures of a wide streamline box girder were investigated via section model wind tunnel tests. The main conclusions are summarized as follows:

1. The inspection vehicle rails and handrails are key factors in the generation of heaving VIV of the wide streamline box girder. After removing the inspection vehicle rails and handrails, the largest heaving VIV amplitude was reduced by 64.2% and 70.3%, respectively. The heaving VIV performance of original deck under $+5^\circ$ attack angle is poor and the optimizations on the VIV countermeasures are need.
2. A sharper wind fairing without changing the distance of the railings can reduce the flow separation at the leading edge of box girder and is favorable to the heaving VIV. The wind fairing in 53° angle can reduce the heaving VIV amplitude by 11.5%, however it cannot satisfy the design requirement.
3. Different from most of previous studies, the countermeasures of inspection vehicle rail and guide vanes have slight influences on attenuating the heaving VIV responses, and the heaving VIV amplitude was reduced by no more than 30.1%.
4. The traffic barriers sealed with plates by regular intervals along the bridge deck can successfully suppress the heaving VIV. The traffic barrier in 41.0% porosity (the interval of plates is 5) can reduce the heaving VIV amplitude by 45.3%, which can satisfy the design requirements. However, it would increase the drag forces and affect driving comfort and the appearance of the bridge negatively.
5. The porosity of handrail has a significant effect on the heaving

VIV. With the application of the simplified circular shape handrails in porosity of 73%, the heaving VIV amplitude was reduced by 65.5%. The validity of this countermeasure was confirmed via the large-scale section model test.

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